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1 PhD Research Milestones

1.1 Key Achievements and Checkpoints

Major Milestones for 4-Year PhD Program

Last Updated: December 22, 2025

1.2 Overview

This document outlines the key milestones for the PhD research on AI-Based Intrusion Detection Systems for Next-Generation Networks. Each milestone represents a significant achievement that marks progress toward the PhD degree.

1.3 Year 1 Milestones

1.3.1 M1: Committee Formation (Month 3)

Target Date: End of Month 3

Description: Dissertation committee formed with 4-5 faculty members, including advisor and external member

Criteria for Success: - Committee members identified and agreed - Initial research direction approved - Meeting schedule established **Deliverables:** - Committee formation documentation - Initial research plan

1.3.2 M2: CNN-LSTM Prototype Working (Month 6)

Target Date: End of Month 6

Description: First hybrid deep learning model implemented and tested

Criteria for Success: - Model architecture implemented in TensorFlow/PyTorch - Training on NSL-KDD and CICIDS2017 complete - Accuracy >95% on test sets - Baseline comparison documented **Deliverables:** - Working code repository - Technical report with results - Performance metrics

1.3.3 M3: All Core Models Implemented (Month 9)

Target Date: End of Month 9

Description: CNN-LSTM, Transformer, Autoencoder, and Ensemble methods all operational

Criteria for Success: - All 4 model types implemented - Comparative evaluation complete - Initial XAI integration (SHAP, LIME) **Deliverables:** - Complete model suite - Comparison report - Visualization dashboard

1.3.4 M4: Comprehensive Exam Passed (Month 11)

Target Date: Month 11

Description: Successfully complete PhD comprehensive examination

Criteria for Success: - Written exam passed (if required) - Oral presentation delivered - Committee approval to advance to candidacy **Deliverables:** - Exam presentation slides - Candidacy status ☐ **CRITICAL MILESTONE** - Required to continue in PhD program

1.3.5 M5: First Paper Submitted (Month 12)

Target Date: End of Month 12

Description: First research paper submitted to workshop or conference

Criteria for Success: - Paper written (6-10 pages) - Results on CNN-LSTM hybrid - Submitted to workshop (e.g., IEEE workshops) or mid-tier conference **Deliverables:** - Submitted manuscript - Submission confirmation

1.4 Year 2 Milestones

1.4.1 M6: Reinforcement Learning Working (Month 15)

Target Date: End of Month 15

Description: Adaptive learning mechanisms implemented and validated

Criteria for Success: - DQN or PPO agent trained successfully - Demonstrated improvement in dynamic threat response - Online learning capability functional **Deliverables:** - RL agent implementation - Adaptation experiments report

1.4.2 M7: Federated Learning Framework Complete (Month 18)

Target Date: End of Month 18

Description: Distributed IDS architecture fully developed

Criteria for Success: - FedAvg implementation working with 10+ clients - Model compression achieved (4-8 $\times$ reduction) - Privacy guarantees (differential privacy) integrated **Deliverables:** - Federated learning codebase - Privacy analysis report - Distributed experiments results

1.4.3 M8: Testbed Operational (Month 21)

Target Date: End of Month 21

Description: Physical hardware testbed deployed and functional

Criteria for Success: - 5 SDN switches + ONOS controller working - 10 Raspberry Pi edge nodes deployed - 50 IoT devices connected - Network monitoring (Grafana, Prometheus) active **Deliverables:** - Testbed documentation - Network topology diagrams - Connectivity test results

1.4.4 M9: Successful Testbed Deployment (Month 24)

Target Date: End of Month 24

Description: IDS deployed on testbed with real-time detection capability

Criteria for Success: - Edge-fog-cloud architecture deployed - Real-time detection latency <100ms - Successful attack simulations (DoS, port scan, web attacks) **Deliverables:** - Deployment guide - Real-world experimental results - Performance benchmark report

1.4.5 M10: Second Major Publication Submitted (Month 24)

Target Date: End of Month 24

Description: Journal paper on distributed IDS submitted

Criteria for Success: - Full paper (8-12 pages for conference, 15-20 for journal) - Comprehensive distributed IDS evaluation - Submitted to top venue (IEEE TIFS, IEEE TMC, or top conference) **Deliverables:** - Journal manuscript - Submission confirmation

1.5 Year 3 Milestones

1.5.1 M11: All Experiments Complete (Month 27)

Target Date: End of Month 27

Description: Extensive evaluation on all datasets finished

Criteria for Success: - Experiments on 5 datasets (NSL-KDD, CICIDS2017, UNSW-NB15, IoT-23, 5G custom) - Cross-dataset validation complete - Ablation studies finished - Statistical significance tests performed **Deliverables:** - Complete experimental results - Statistical analysis report - Comparison tables and figures

1.5.2 M12: 5G/IoT Validation Complete (Month 30)

Target Date: End of Month 30

Description: Next-generation network specific validation completed

Criteria for Success: - Custom 5G dataset collected (1M+ flows) - IoT-23 malware detection >95% accuracy - Edge computing performance validated (<100ms latency on Raspberry Pi) **Deliverables:** - 5G dataset and documentation - IoT malware detection report - Edge performance analysis

1.5.3 M13: Performance Targets Met (Month 33)

Target Date: End of Month 33

Description: All performance optimization goals achieved

Criteria for Success: - Detection latency <100ms (mean), <200ms (p99) - Throughput >10K packets/second - Accuracy >98%, Precision >97%, Recall >98%, F1 >97% - False positive rate <1% **Deliverables:** - Optimized model artifacts - Performance benchmark report - Optimization documentation

1.5.4 M14: Framework Publicly Released (Month 36)

Target Date: End of Month 36

Description: Open-source AI-IDS framework published on GitHub

Criteria for Success: - GitHub repository created and populated - Comprehensive README and documentation - Docker images published - Example notebooks and tutorials - Community engagement started (stars, forks, issues) **Deliverables:** - GitHub repository: [github.com/\[username\]/ai-ids-ngn](https://github.com/[username]/ai-ids-ngn) - Documentation website - Tutorial videos □ **MAJOR MILESTONE** - Research contribution to community

1.5.5 M15: 3+ Publications Submitted (Month 36)

Target Date: End of Month 36

Description: Multiple research papers submitted to top venues

Criteria for Success: - At least 3 papers submitted (mix of conference and journal) - At least 1 paper accepted/published - Papers cover different aspects (architecture, distributed, explainability) **Deliverables:** - Submitted manuscripts - Acceptance letters (for accepted papers) - Preprints on arXiv

1.6 Year 4 Milestones

1.6.1 M16: User Studies Complete (Month 39)

Target Date: End of Month 39

Description: Explainability evaluation with human subjects finished

Criteria for Success: - 10-20 security analysts recruited - Questionnaire administered - User trust score >4.0/5.0 - Qualitative feedback collected and analyzed **Deliverables:** - User study report - Survey results and analysis - Recommendations for UX improvement

1.6.2 M17: Thesis Draft Complete (Month 42)

Target Date: End of Month 42

Description: Complete dissertation written and ready for committee review

Criteria for Success: - All chapters written (7 chapters, 200-300 pages) - All figures and tables created - References complete (100+ citations) - Formatted according to university guidelines **Deliverables:** - Complete thesis draft (PDF) - Chapter-by-chapter breakdown □ **MAJOR MILESTONE** - Thesis ready for review

1.6.3 M18: Thesis Ready for Defense (Month 45)

Target Date: End of Month 45

Description: Revised thesis incorporating committee feedback

Criteria for Success: - All committee comments addressed - Defense presentation prepared (45-60 minutes) - Defense date scheduled - Committee approval to defend

Deliverables: - Final thesis draft - Defense presentation slides - Demo materials

1.6.4 M19: Defense Passed (Month 46)

Target Date: Month 46

Description: Successfully defend PhD dissertation

Criteria for Success: - Public presentation delivered - Committee questions answered satisfactorily - Pass vote from committee (unanimous or majority) - Minor revisions accepted (if any)

Deliverables: - Defense certificate - Committee signatures - Passed candidacy □ **CRITICAL MILESTONE** - Near completion!

1.6.5 M20: Final Dissertation Submitted (Month 48)

Target Date: Month 48

Description: Final dissertation submitted to university

Criteria for Success: - All defense revisions incorporated - Final proofreading complete - Format check passed - Submitted to graduate school - University approval received

Deliverables: - Final dissertation (PDF) - Submission receipt - Graduation clearance

1.6.6 M21: PhD DEGREE AWARDED (Month 48)

Target Date: End of Month 48

Description: GRADUATION - DOCTOR OF PHILOSOPHY AWARDED! □

Criteria for Success: - All degree requirements completed - Dissertation approved and archived - Graduation ceremony attended - Diploma received

Deliverables: - **PhD DEGREE** - Published dissertation - 5-10 peer-reviewed publications - Open-source framework with community adoption □ **ULTIMATE MILESTONE** - PhD Completed!

1.7 Summary Table

Milestone	Month	Type	Critical?
M1: Committee Formation	3	Administrative	□
M2: CNN-LSTM Prototype	6	Technical	

Milestone	Month	Type	Critical?
M3: All Core Models	9	Technical	
M4: Comprehensive Exam	11	Academic	☐ CRITICAL
M5: First Paper Submitted	12	Publication	
M6: RL Working	15	Technical	
M7: Federated Framework	18	Technical	☐
M8: Testbed Operational	21	Infrastructure	☐
M9: Testbed Deployment	24	Technical	
M10: Second Paper Submitted	24	Publication	
M11: All Experiments	27	Research	
M12: 5G/IoT Validation	30	Research	
M13: Performance Targets	33	Technical	☐
M14: Framework Released	36	Impact	☐ MAJOR
M15: 3+ Papers Submitted	36	Publication	
M16: User Studies	39	Research	
M17: Thesis Draft	42	Academic	☐ MAJOR
M18: Defense Ready	45	Academic	
M19: Defense Passed	46	Academic	☐ CRITICAL
M20: Final Submission	48	Administrative	
M21: PhD AWARDED	48	COMPLETION	☐ ULTIMATE

1.8 Tracking and Reporting

1.8.1 Progress Tracking

- Monthly progress reports to advisor
- Quarterly committee meetings
- Annual comprehensive review

1.8.2 Milestone Documentation

For each milestone: 1. **Pre-Milestone:** Plan and prepare (1-2 weeks before) 2. **Milestone Event:** Execute and achieve 3. **Post-Milestone:** Document and reflect (1 week after)

1.8.3 Risk Management

- Buffer time built into schedule (2-4 weeks per phase)
- Alternative plans for critical milestones
- Regular risk assessments with advisor

1.9 Conclusion

These 21 milestones provide a clear roadmap from PhD enrollment to graduation. Regular achievement of these milestones ensures steady progress and on-time completion of the doctoral degree. The mix of technical, academic, and publication milestones ensures comprehensive development as a researcher.

Keys to Success: - ☐ Stay on track with quarterly goals - ☐ Regular communication with advisor and committee - ☐ Document progress continuously - ☐ Celebrate achievements along the way!

End Goal: Dr. [Your Name], PhD in Computer Science - Expert in AI-Based Network Security! ☐

References: See `gantt_chart.md` for detailed timeline and `../resources/budget.md` for resource allocation per milestone.